STAT 238 Final Project Proposal

Bayesian Optimization for Expensive Black-Box ML Pipelines

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Overview

Hyperparameter tuning and pipeline configuration are unavoidable steps in applied machine learning, yet they are often handled carelessly: grid search wastes evaluations on uninformative regions, and random search ignores everything learned from past trials. Bayesian Optimization (BO) offers a principled alternative. The key idea, described in detail in Frazier (2018), is to maintain a probabilistic surrogate model of the objective function and use it to decide where to evaluate next, trading off between exploring uncertain regions and exploiting regions that look promising. Because the surrogate is cheap to query, this strategy can find good configurations in far fewer evaluations than either grid or random search. This advantage is most significant when each evaluation of the true objective is expensive, for example when training a neural network or running inference through a large language model.

This project implements BO using the Gaussian Process regression framework developed in class, applies it to two real pipelines drawn from concurrent research projects, and compares its sample efficiency against random search and grid search on both tasks. A small simulation study on a known test function is included first to verify the implementation before moving to the real-data experiments.

Experiment 1: Optimizing a Zero-Shot Content Classification Pipeline (Upworthy)

A concurrent project in STAT 230A studies how linguistic features of news headlines predict click-through rate (CTR) using the Upworthy Research Archive, a dataset of 22,666 headline packages from A/B tests conducted between 2013 and 2015. A key step in that analysis is assigning each headline to a content category (e.g., politics, science, lifestyle, health) so that category-by-feature interaction terms can be estimated. The categories are not observed and must be inferred via zero-shot classification using `facebook/bart-large-mnli`.

The classification pipeline has several discrete and continuous choices that are not differentiable: the candidate label set, the number of categories K, and the confidence threshold below which a package is left unassigned. The quality of a given configuration is measured by how well the inferred categories separate the log-CTR distributions across groups, formalized as the F-statistic from a one-way ANOVA of log(CTR) on the inferred category labels. A higher F-statistic means the categories are more informative for the downstream regression.

This is a genuine black-box problem: the objective is not differentiable through the classifier plus the F-statistic, and each evaluation requires running inference over the full dataset, taking roughly 30 seconds per configuration. The plan is to run 60 to 80 BO evaluations and compare the best configuration found against random search and a coarse grid search at matched evaluation budgets. The winning category labels will feed directly into the 230A regression analysis.

Experiment 2: Tuning Neural Collaborative Filtering (MovieLens)

A concurrent project in CS 289A compares matrix factorization, neural collaborative filtering (NCF), and a two-stage ranker on the MovieLens 1M dataset to study how recommendation quality degrades as per-user interaction density decreases. The NCF model and the ranking MLP each have hyperparameters (embedding dimension, MLP hidden layer sizes, learning rate, L2 regularization strength) that must be set before the sparsity analysis can be run fairly. Choosing these by hand or grid search introduces arbitrary choices that could confound the comparison.

BO will be used to tune the NCF hyperparameters with validation NDCG@10 as the objective. Training one NCF configuration on MovieLens 1M takes roughly 3 to 5 minutes on an A4000 GPU, putting this squarely in the regime where sample efficiency matters most. The plan is to run 25 to 30 BO evaluations on the `jsteinhardt` GPU cluster and compare the convergence curve against random search and grid search at the same budget. The best configuration found by BO will be used as the NCF baseline in the CS 289A project.

Methods

Gaussian Process Surrogate

The surrogate model is a Gaussian Process placed over the objective function $f : \mathcal{X} \to \mathbb{R}$, where $\mathcal{X}$ is the hyperparameter space. Following the GP regression setup from Lecture 23, noisy observations $y_i = f(x_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ are assumed. Given n observations $(x_1, y_1), \ldots, (x_n, y_n)$, the GP posterior at a new point x is Gaussian with posterior mean and variance:

$$\mu(x) = k^T \big(K + \sigma^2 I_n\big)^{-1} y, \qquad \sigma^2(x) = K(x, x) - k^T \big(K + \sigma^2 I_n\big)^{-1} k,$$

where K is the $n \times n$ matrix with (i, j) entry $K(x_i, x_j)$, and k is the $n \times 1$ vector with i-th entry $K(x_i, x)$. These are the standard GP regression formulas from Lecture 23.

For the kernel, the squared exponential (SE) kernel will be used,

$$K(x, x') = \alpha^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right),$$

with output scale α and lengthscale ℓ. This kernel was introduced in the course lab on GP regression and is the standard choice in the BO literature. The kernel hyperparameters α, ℓ, and the noise variance σ^2 are estimated at each iteration by maximizing the marginal likelihood of the observed data, following the approach in Section 3.1 of Lecture 23:

$$y \mid \alpha, \ell, \sigma^2 \;\sim\; \mathcal{N}\Big(0,\; K + \sigma^2 I_n\Big).$$

Acquisition Function

At each iteration, BO selects the next configuration by maximizing the Expected Improvement (EI) acquisition function over $\mathcal{X}$. EI quantifies, in expectation, how much the objective at x would improve over the best value observed so far, $f^* = \max_i y_i$. Because the GP posterior at x is Gaussian with mean $\mu(x)$ and standard deviation $\sigma(x)$, EI has the closed form:

$$\alpha_{\mathrm{EI}}(x) = \big(\mu(x) - f^*\big)\,\Phi(Z) \;+\; \sigma(x)\,\phi(Z), \qquad Z = \frac{\mu(x) - f^*}{\sigma(x)},$$

where Φ and ϕ are the standard normal CDF and PDF. The term $\mu(x) - f^*$ drives exploitation toward regions with high predicted values, while $\sigma(x)$ drives exploration toward uncertain regions. EI balances these two goals without requiring additional tuning. The acquisition function is maximized numerically at each step using `BoTorch`.

Algorithm

The full BO loop is:

1. Evaluate f at $n_0 = 5$ to 10 random configurations to initialize.
2. Fit the GP surrogate to all observed (x_i, y_i) pairs by maximizing the marginal likelihood.
3. Maximize $\alpha_{\mathrm{EI}}(x)$ over $\mathcal{X}$ to obtain the next candidate x^*.
4. Evaluate $f(x^*)$, append to observations, and return to step 2.
5. Repeat until the evaluation budget is exhausted.

Both experiments use the same loop; the only difference is the black-box objective evaluated at step 4.

Simulation Study

Before applying BO to the two real-data experiments, the implementation will be verified on the Branin function, a standard two-dimensional BO benchmark with a known global minimum. This serves as a sanity check and allows for visualizing how the GP surrogate and the EI acquisition function evolve across iterations, something that is harder to illustrate in the higher-dimensional real-data settings.

Comparison and Deliverables

For each experiment, the notebook will report convergence curves showing the best observed objective as a function of the number of evaluations, for BO, random search, and grid search. The GP surrogate and acquisition function will also be visualized at selected iterations of the simulation study to illustrate how the model updates its beliefs across iterations. The presentation will cover the BO methodology, the two experimental setups, and a discussion of where BO provides the clearest efficiency gains and where the overhead of fitting the GP and optimizing the acquisition function is less justified.

References

The following sources will guide the implementation and writeup.

- Frazier, P. I. (2018). A tutorial on Bayesian Optimization. *arXiv:1807.02811*.
- Snoek, J., Larochelle, H., and Adams, R. P. (2012). Practical Bayesian Optimization of Machine Learning Algorithms. *NeurIPS 2012*.
- Hennig, P., Osborne, M. A., and Kersting, H. P. (2022). *Probabilistic Numerics: Computation as Machine Learning*. Cambridge University Press.

Datasets

Both datasets are already in use in the concurrent projects noted above. The Upworthy Research Archive is publicly available at `osf.io/jd64p/`. The MovieLens 1M dataset is available through the GroupLens repository at `grouplens.org/datasets/movielens/1m/`. Using shared datasets across courses is disclosed here; the analyses in each course are entirely distinct.